International vulnerability of inflation

XIV WTSE - Workshop in Time Series Econometrics
April 11, 2024

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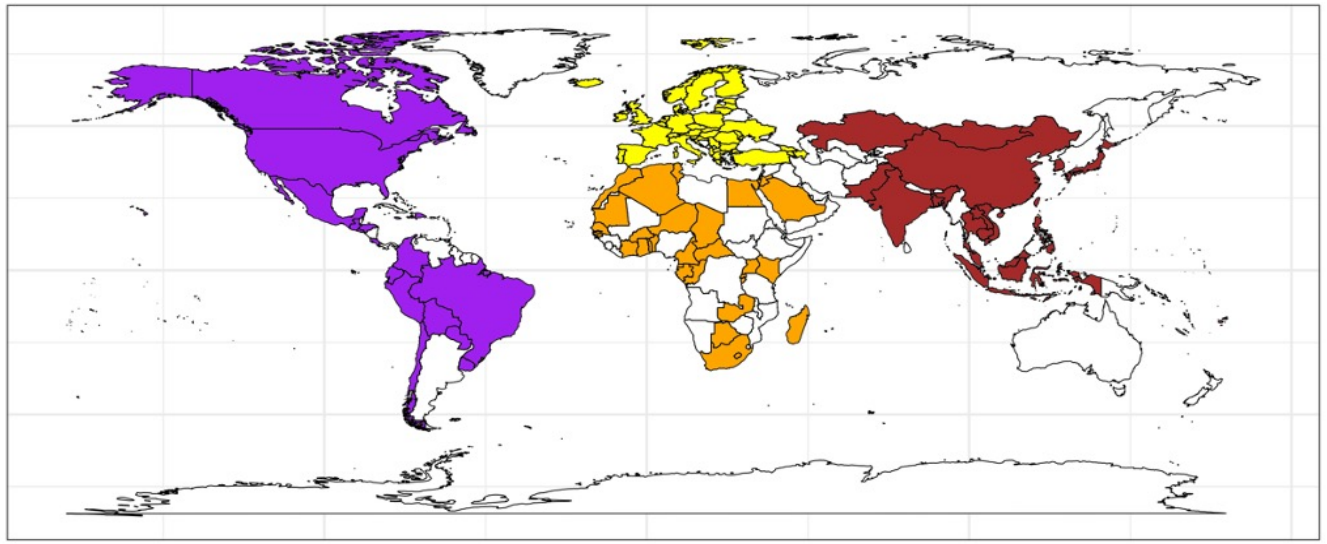
This paper

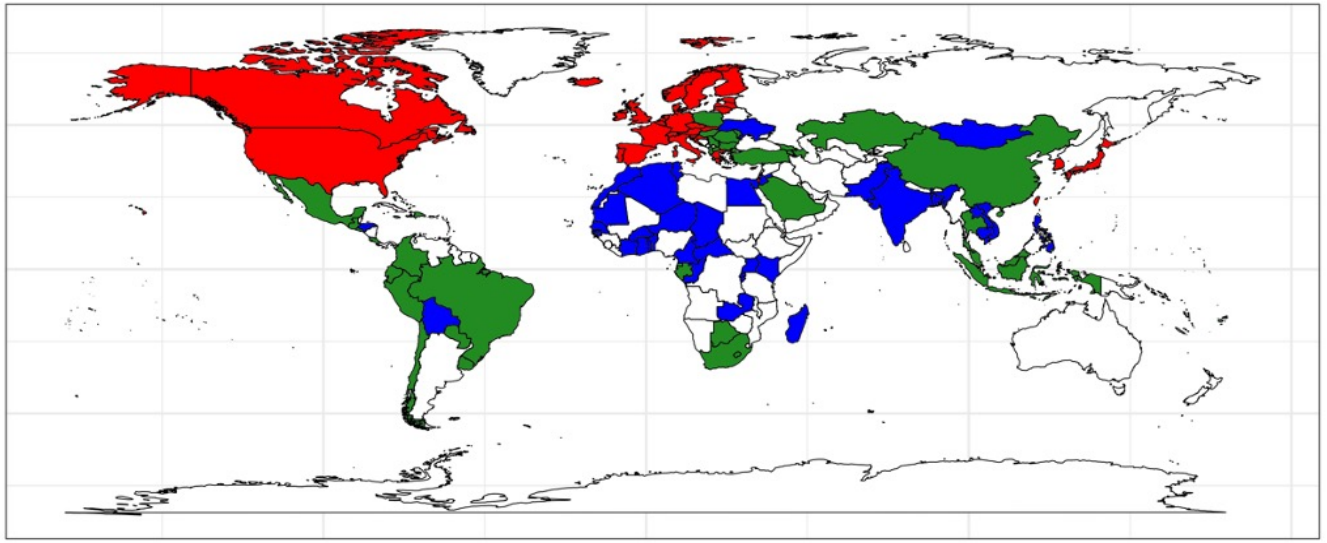
Constructs inflation densities for 115 countries (1999-2022) based on international factors that can be linked to the geographical region and the income level of each country.

Geographical regions



Countries (115): from 1999 to 2022 Africa (33) America (21) Asia & Oceania(22) Europe (39)

Income level



Countries (115): from 1999 to 2022 Advanced Economies (35) EMDEs (42) Low income EMDEs (38)

Multi-level Dynamic Factor Model

- ML-DFM:

$$Y_t = \Lambda F_t + \varepsilon_t,$$

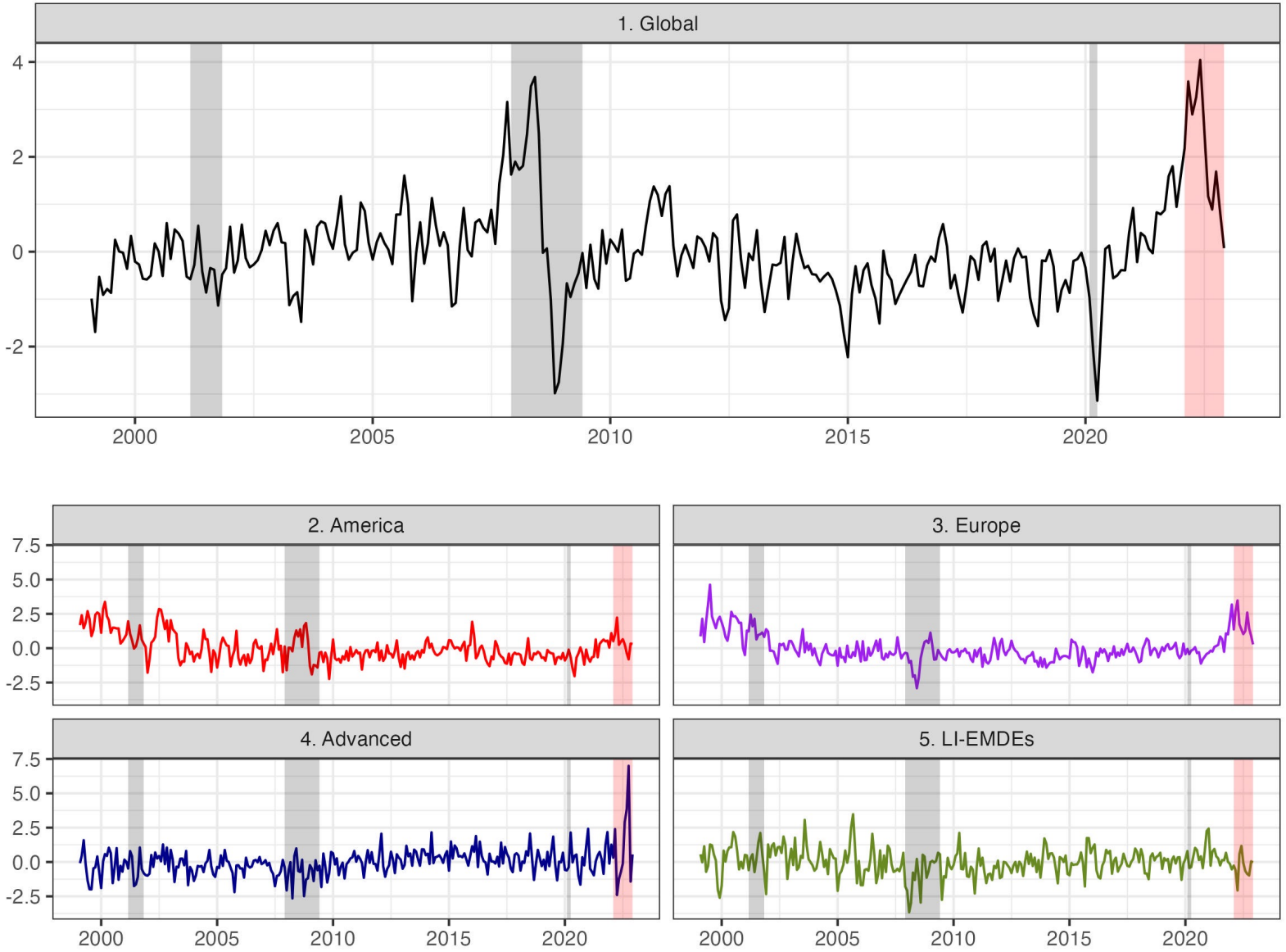
Advanced economies in Europe	Y_{1t}		λ_{11}	λ_{12}	0	λ_{14}	0	
HMI-EMDE in Europe	Y_{2t}		λ_{21}	λ_{22}	0	0	0	
LI-EMDE in Europe	Y_{3t}		λ_{31}	λ_{32}	0	0	λ_{35}	F_{gt} Global factor
Advanced economies in America	Y_{4t}		λ_{41}	0	λ_{43}	λ_{44}	0	F_{Eurt} Europe factor
HMI-EMDE in America	Y_{5t}	$=$	λ_{51}	0	λ_{53}	0	0	F_{Amet} America factor
LI-EMDE in America	Y_{6t}		λ_{61}	0	λ_{63}	0	λ_{65}	F_{Devt} Developed factor
Advanced in Asia, O. and Africa	Y_{7t}		λ_{71}	0	0	λ_{74}	0	F_{LIt} LI-EMDE factor
HMI-EMDE in Asia, O. and Africa	Y_{8t}		λ_{81}	0	0	0	0	
LI-EMDE in Asia, O. and Africa	Y_{9t}		λ_{91}	0	0	0	λ_{95}	

- Sequential Least Squares estimation (see, Breitung and Eickmeier, 2016; Rodríguez-Caballero and Caporin, 2019).

Results I

The estimated factors capture past shocks:

- GFC (2007M12-2009M06).
- Covid-19 (2020M2-2020M4).
- Russia-Ukraine war (2022M2-)



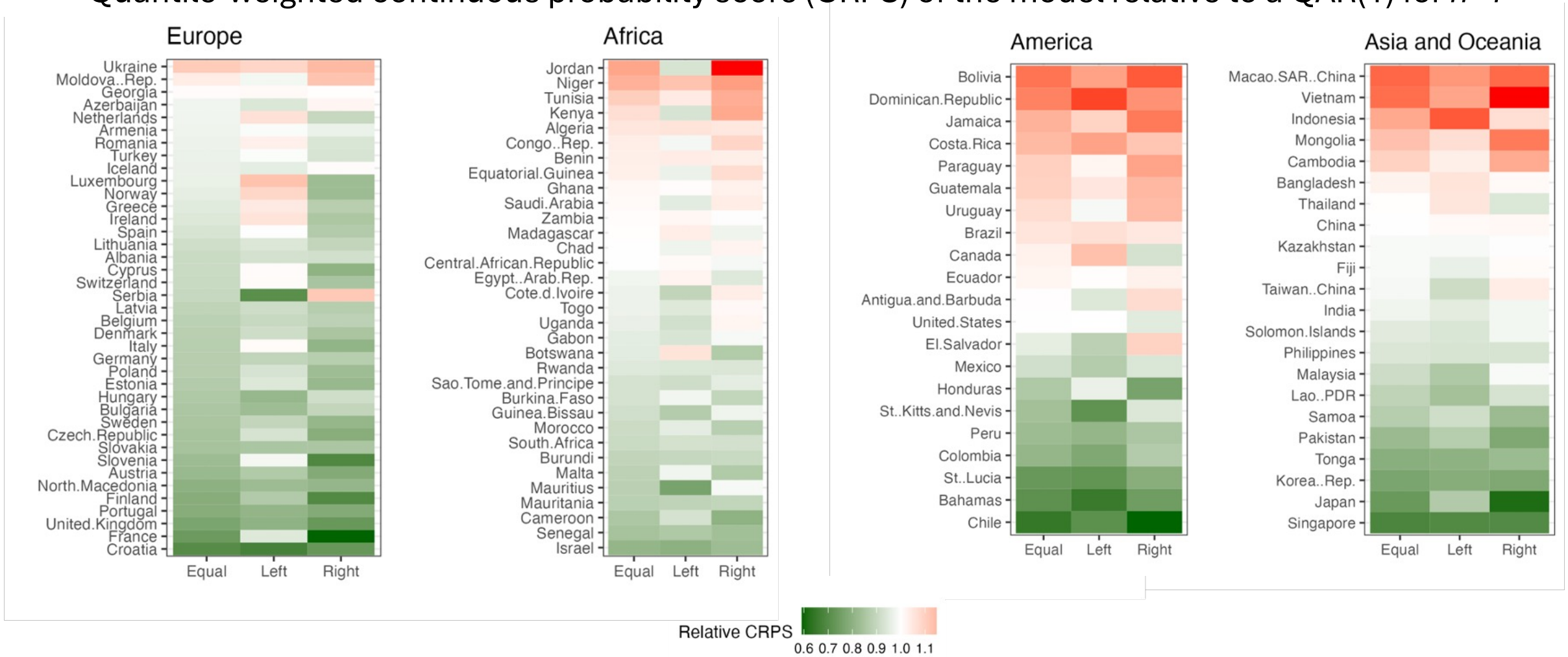
Note: Standardized factors. Time span from February 1999 to December 2022. The gray shaded area represents recessions for United States of America identified by the National Bureau of Economic Research and the red shaded area denotes the Russia-Ukraine war.

Results II

$$q_{\tau^*}\left(y_{it+h}|y_{it},F_t\right)=\mu(\tau^*,i,h)+\phi(\tau^*,i,h)y_{it}+\sum_{k=1}^{r}\beta_k(\tau^*,i,h)F_{kt},$$

- Forecasting exercise for January 2020 to December 2022.

Quantile-weighted continuous probability score (CRPS) of the model relative to a QAR(1) for *h=1*



Conclusions and further research

- In this paper, we propose a novel methodology to put together two strands in the literature (regional and income-level factors) by modelling domestic inflation density and by considering international movements of inflation.
- On-going research: stress scenarios for inflation (González-Rivera, Maldonado and Ruiz, 2019; González-Rivera, Rodriguez-Caballero, and Ruiz, 2024; Maldonado and Ruiz, 2021).

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